

Write the equation in its equivalent form.

1. $\log_3 x = 10$

Write the equation in its equivalent logarithmic form.

2. $6^4 = x$

Use properties of logarithms to condense the expression as single logarithm whose coefficient is 1. Where possible, evaluate the logarithmic expression.

3. $3 \log_y 4 + \log_y 3$

4. $6 \ln x - \frac{1}{4} \ln y$

Use the properties of logarithms to expand the logarithmic expression. Write the expression as a single logarithm whose coefficient is 1.

5. $\log_b(xy^6)$

6. $\log_5 \left(\frac{7x}{2} \right)$